UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	OCTOBER 2026
TIME	2 HOURS

NSTRUCTIONS TO CANDIDATES

1. This question paper consists of 5 questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) a one – page Appendix 1
   iii) an Answer Booklet - provided by the Faculty
   iv) a graph paper - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages

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QUESTION 1

a) Solve the inequality $3x^2 - 9x - 12 > 0$.

(5 marks)

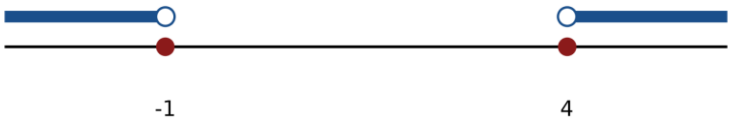


Figure: critical values.

b) Solve the inequality $|4 - x| < 2$. Express the answer in interval notation and on a number line.

(5 marks)



Figure: solution on a number line.

c) Solve the equation $|2x + 6| = 4$.

(4 marks)

QUESTION 2

a) Given $f(x) = 2x + 2$ and $g(x) = 5 - 4x$, find $(f + g)(x)$, $(f - g)(x)$ and $(f\,g)(x)$. State the domain of each.

(4 marks)

b) Given $f(x) = x + 1$ and $g(x) = 2/(x + 4)$, find $(g \circ f)(x)$ and state its domain.

(5 marks)

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c) Write the equation of the circle with centre (5, -3) and radius 5. Hence write its general equation and sketch the circle.

(5 marks)

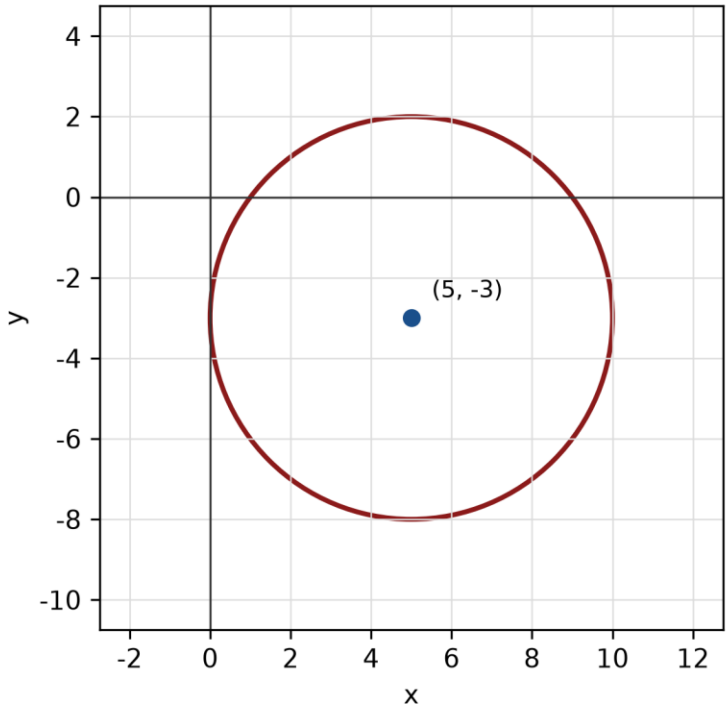


Figure: circle.

QUESTION 3

a) Given $z_1 = -6 + 6i$ and $z_2 = -6 + 6i$, find $z_1 + z_2$ in the form a + bi.

(4 marks)

b) Evaluate i^28 and simplify completely.

(3 marks)

c) Solve the exponential equation 3^(-2x - 5) = 3^(x - 2).

(5 marks)

QUESTION 4

a) Solve the system of equations simultaneously:
x - 3y - 3z = 0
3x - y - 4z = 4
-x - y - z = 4

(5 marks)

b) Solve the simultaneous equations
2x - 2y = -12
-4x + 3y = 19.

(4 marks)

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c) Write $x^2 - 4x + y^2 + 6y - 12 = 0$ in standard form. Hence state the centre and radius and sketch the circle.

(5 marks)

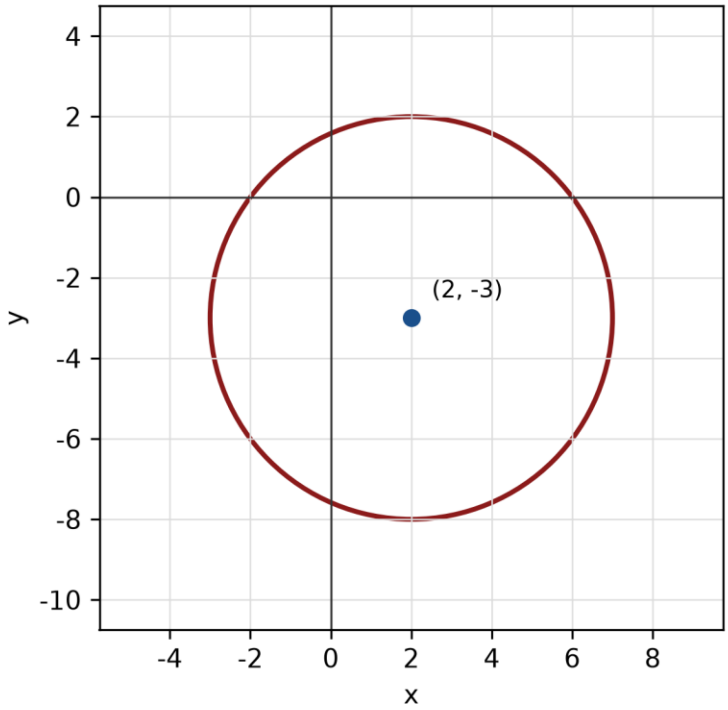


Figure: circle.

QUESTION 5

a) Prove the identity $(1 + \cos x)/\sin x + \sin x/(1 + \cos x) = 2 \csc x$.

(5 marks)

b) A circle has radius 11 cm. Find the length of the arc subtended by a central angle of 150°.

(4 marks)

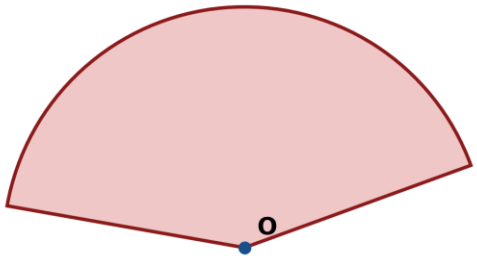


Figure: arc and sector.

c) Solve the triangle with sides a = 10 cm, b = 24 cm and c = 26 cm. Find all three angles.
(5 marks)

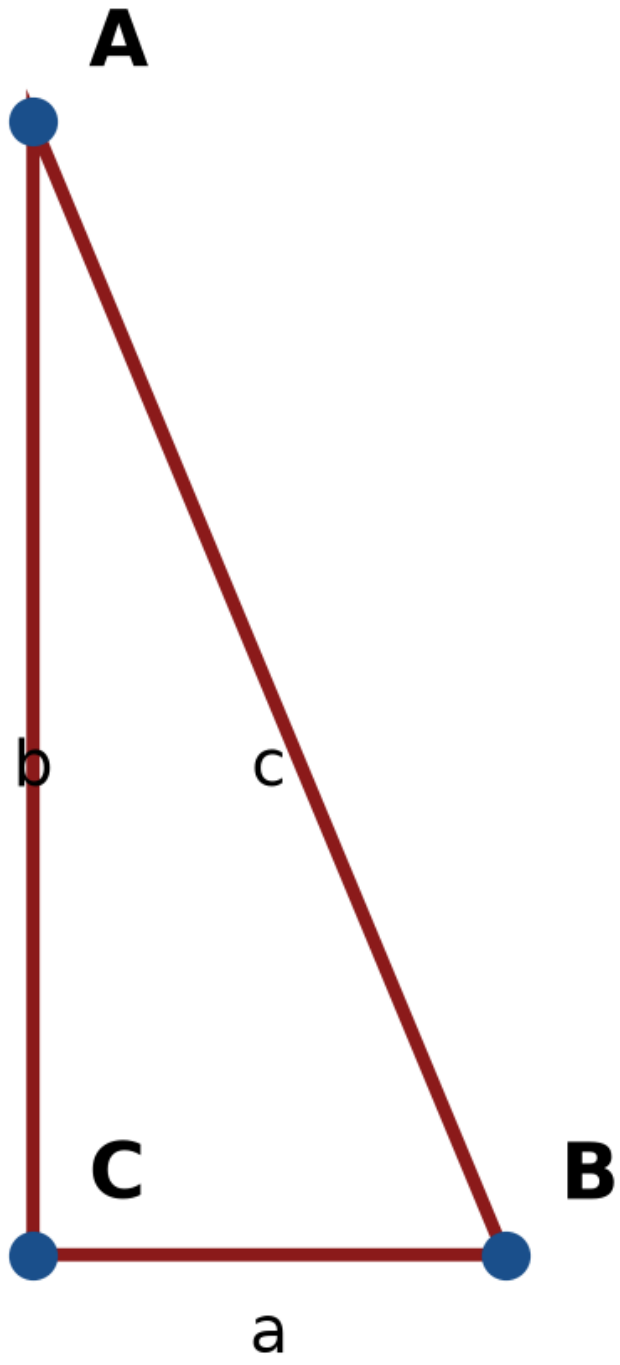


Figure: right triangle.

END OF QUESTION PAPER

APPENDIX 1

Formula sheet (as supplied with the paper).

$\sin^2 A + \cos^2 A = 1$
$\sin(A + B) = \sin A \cos B + \cos A \sin B$
$\cos(A + B) = \cos A \cos B - \sin A \sin B$
$\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \tan B)$
$\cos 2A = \cos^2 A - \sin^2 A$
$\cos 2A = 1 - 2\sin^2 A = 2\cos^2 A - 1$
$\sin 2A = 2 \sin A \cos A$
$\tan^2 A + 1 = \sec^2 A$
$\tan 2A = 2 \tan A / (1 - \tan^2 A)$
Arc length, $s = r\theta$
Area of sector = $\frac{1}{2}r^2\theta$
$a / \sin A = b / \sin B = c / \sin C$
$a^2 = b^2 + c^2 - 2bc \cos A$
Area of triangle = $\frac{1}{2}ab \sin C$

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